

Pooled Correlations Hide the Structure: Neural-Collapse Coordinates and Held-Out Accuracy Within Training Conditions

David Lindgreen

October 9, 2026

Version 0.1.0 Preprint, not peer reviewed

Abstract

Neural collapse describes four geometric regularities (NC1–NC4) of a classifier’s last-layer features, often observed late in training [1]. We tested, in a small and fully inspectable setting, whether they continue to strengthen after interpolation and whether their strength tracks held-out accuracy: a $64 \rightarrow 64 \rightarrow 9 \rightarrow 10$ ReLU network on the official writer-disjoint UCI Optical Digits split [2], ten seeds in each of three training conditions (clean, long-tailed, 20% label noise). Two of four preregistered directional gates passed; the re-run of the whole study reproduces the frozen results (largest numeric difference 3.4×10^{-6} , identical gates). The original report quoted a descriptive Spearman association of -0.518 between final NC1 and test accuracy over all 30 runs. We show that this pooled figure is produced by the three conditions, not by NC1: inside each condition the association is $+0.04$, -0.04 and -0.09 , and the pooled within-condition association is -0.03 (95% seed-bootstrap interval -0.49 to $+0.42$). The same holds for NC3 (pooled -0.75 , within $+0.05$). Only NC4, the disagreement between the network and nearest-class-centre decisions, is associated with accuracy within conditions (-0.49 ; -0.80 to -0.04), consistently across all three. This is an instance of Simpson’s paradox [3] in a two-variable summary.

1 Question

After a small network reaches zero training error, does its hidden representation continue toward neural-collapse geometry, and does stronger collapse consistently accompany better generalization to unseen writers? The setting is deliberately small: 3,823 training and 1,797 test examples from disjoint writers, a multinomial logistic reference, and ten algorithmic seeds per condition (seeds are replications of the optimizer, not samples of people).

2 Frozen design and results

The three conditions were the clean training set, a deterministic long tail (class c keeps $\lfloor n_c 10^{-c/9} \rfloor$ rows, at least 25; 1,559 rows) and exactly 20% symmetric label noise per class (761 changed labels). Models were trained for 600 Adam epochs. The collapse coordinates are NC1 (within-class variability relative to between-class, lower is more collapsed), NC2 (deviation of centred class means from the simplex equiangular frame), NC3 (classifier–class-mean misalignment) and NC4 (disagreement with nearest-class-centre decisions), computed in the full nine-dimensional hidden space. Clean-condition MLP test accuracy had a median of 96.27% (95.88–96.94%), long tail 92.29%, label noise

Preregistered gate (at least 8 of 10 seeds)	Result	Decision
Clean: final NC1 below first-zero-error NC1	7/10	not passed
Clean: final NC2 below first-zero-error NC2	7/10	not passed
Clean: final NC4 no greater than first-zero-error NC4	10/10	passed
Long tail: final NC2 worse than clean (paired)	10/10	passed

Table 1: Frozen directional gates; two of four pass.

81.64%; the logistic reference reached 94.77%, 91.88% and 93.60%. The frozen report quoted descriptive pooled Spearman associations of -0.518 (NC1) and -0.026 (NC2) with test accuracy over the 30 MLP runs and reported no p -values, since seeds are not independent samples of a population. We re-ran the entire study from the pinned dataset: every number reproduces (largest absolute difference 3.4×10^{-6} in a class-angle Gram entry), as do all gate counts and both pooled correlations.

3 Post-hoc: associations within conditions

The three conditions differ by more than ten accuracy points, so any metric that also differs between conditions will correlate with accuracy in a pooled sample whether or not it tracks accuracy among comparable runs. We therefore recomputed each coordinate’s Spearman association with final test accuracy (i) pooled, (ii) inside each condition and (iii) pooled within condition (ranks computed inside each condition), with percentile intervals from a 5,000-resample bootstrap over seeds within condition. This analysis was not preregistered and all of it is reported. For NC1 and NC3 the pooled

	Pooled	Within (pooled)	95% interval	Clean	Long tail	Label noise
NC1	-0.518	-0.031	-0.49 to 0.42	$+0.04$	-0.04	-0.09
NC2	-0.026	-0.310	-0.63 to 0.12	-0.56	-0.24	-0.13
NC3	-0.750	$+0.052$	-0.37 to 0.46	-0.16	$+0.21$	$+0.13$
NC4	-0.622	-0.486	-0.80 to -0.04	-0.56	-0.44	-0.45

Table 2: Spearman association between each final collapse coordinate and test accuracy; each condition has 10 seeds.

association disappears within conditions: the apparent link between “more collapsed” and “more accurate” is the difference between conditions. NC2 shows a weak, imprecise negative within-condition association (largest in the clean condition), even though its pooled value is near zero because the conditions pull in opposite directions. NC4 stands out: its within-condition association is moderately negative in every condition and its pooled-within interval excludes zero, though each single-condition interval does not.

4 Discussion

The frozen conclusion, that a single collapse coordinate is not a general certificate of generalization, is supported and, if anything, sharpened: NC1 and NC3 are condition markers rather than predictors of accuracy among comparable runs, and the one coordinate that tracks accuracy within conditions is the decision-level one (NC4). The pooled NC1 figure in the original report should be read as a between-condition association. The within-condition estimates are imprecise (ten seeds

per condition, intervals wide) and describe seed-to-seed variation in one small architecture, dataset and optimizer; they neither establish nor rule out a causal role for collapse.

5 Limitations

One architecture, one dataset, three hand-built conditions, ten seeds each; collapse coordinates measured at the final epoch only; the bootstrap resamples seeds, which are not draws from a population of tasks; and the within-condition analysis is exploratory.

Reproducibility and AI assistance

Code, frozen results, the within-condition audit and tests:

<https://github.com/lindgreendavid/neural-geometry-lab>

Interactive laboratory:

<https://lindgreendavid.github.io/neural-geometry-lab/>

The analysis code, tests and drafts of this text were produced with substantial assistance from an AI system (Claude, Anthropic); the author is responsible for the content.

References

- [1] V. Pappayan, X. Y. Han, D. L. Donoho, Prevalence of neural collapse during the terminal phase of deep learning training, *Proc. Natl. Acad. Sci. USA* **117**, 24652–24663 (2020).
- [2] E. Alpaydin and C. Kaynak, Optical Recognition of Handwritten Digits, UCI Machine Learning Repository (1998), doi:10.24432/C50P49.
- [3] E. H. Simpson, The interpretation of interaction in contingency tables, *J. R. Stat. Soc. B* **13**, 238–241 (1951).